MIL-I-7186
30 APRIL 1951
Superseding
Air Force 21304
16 May 1949

MILITARY SPECIFICATION

INDICATOR, TACHOMETER, ELECTRIC, SENSITIVE, TYPE E-30

This specification was approved by the Departments of the Army, the Navy, and the Air Force for use of procurement services of the respective Departments.

1. SCOPE

1.1 This specification covers one type of hermetically sealed remote indicating tachometer indicator, designated type E-30, for use at the flight engineer's station in reciprocating engine aircraft, to indicate engine speeds in the range of 0 to 4,000 rpm.

2. APPLICABLE SPECIFICATIONS AND DRAWINGS

2.1 The following publications, of the issue in effect on date of invitation for bids, shall form a part of this specification to the extent specified herein:

2.1.1 Specifications.-

Military

MIL-I-7098	Indicators, Tachometer, Electric, Magnetic Drag Type, Aircraft, General Specification for

2.1.2 Drawings.-

Air Force-Navy Aeronautical Standard Drawings

AND10066	Receptacle - Electrical Connector, Integral Mounting
AND10412	Case - 2-Inch Size Instrument, Standard Dimensions For

(Copies of this specification and copies of other publications referenced herein or required for Government procurement, and the Index of Military Aeronautical (AN or MIL) Standards, may be obtained upon application to the Commanding General, Air Materiel Command, Wright-Patterson Air Force Base, Dayton, Ohio; or the Commanding Officer, U. S. Naval Air Station, Johnsville, Pennsylvania.)

3. REQUIREMENTS

3.1 General.- The requirements specified in Specification MIL-I-7098 are applicable as requirements of this specification. Additional requirements shall be as specified herein.

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3.1.1 In the event the requirements of the general specification and this specification conflict, the requirements of this specification shall govern.

3.2 Design and Construction.-

3.2.1 Case.- The case shall conform to figure 1 and shall be hermetically sealed.

3.2.1.1 The distance between the outer surface of the cover glass and the front edge of the case shall be the minimum practicable and shall not exceed 1/32 inch.

3.2.1.2 The portion of the instrument case that is visible from the front of the instrument panel shall be finished in durable dull black. The finish on the remainder of the case is optional provided that the finish used will in no way prevent compliance with the requirements of this specification.

3.2.2 Electrical Receptacle.- Each indicator shall be provided with a three-pin electrical receptacle conforming to AND10066D-10SL-3P, located on the vertical and horizontal center line of the indicator with the polarizing key of the receptacle at the top.

3.2.3 Pointers.- Two concentrically mounted pointers shall be provided, both reading in rpm with reference to a single scale graduated uniformly as shown on figure 1, as follows: A long pointer making one revolution for each 1,000 rpm and a short pointer making 1/10 of a revolution for each 1,000 rpm. The pointers shall be so installed that the short pointer is nearest the dial.

3.2.3.1 The pointers shall conform to figure 1.

3.2.3.2 The length of the long pointer shall be such that the tip will extend into the scale a distance equivalent to 1/3 to 2/3 of the length of the shortest graduation.

3.2.3.3 Indication.- When driven at 1/2 crankshaft speed, the tachometer indicator shall indicate 0 to 4,000 rpm of the engine crankshaft in four revolutions of the long pointer and 0.40 revolution of the short pointer.

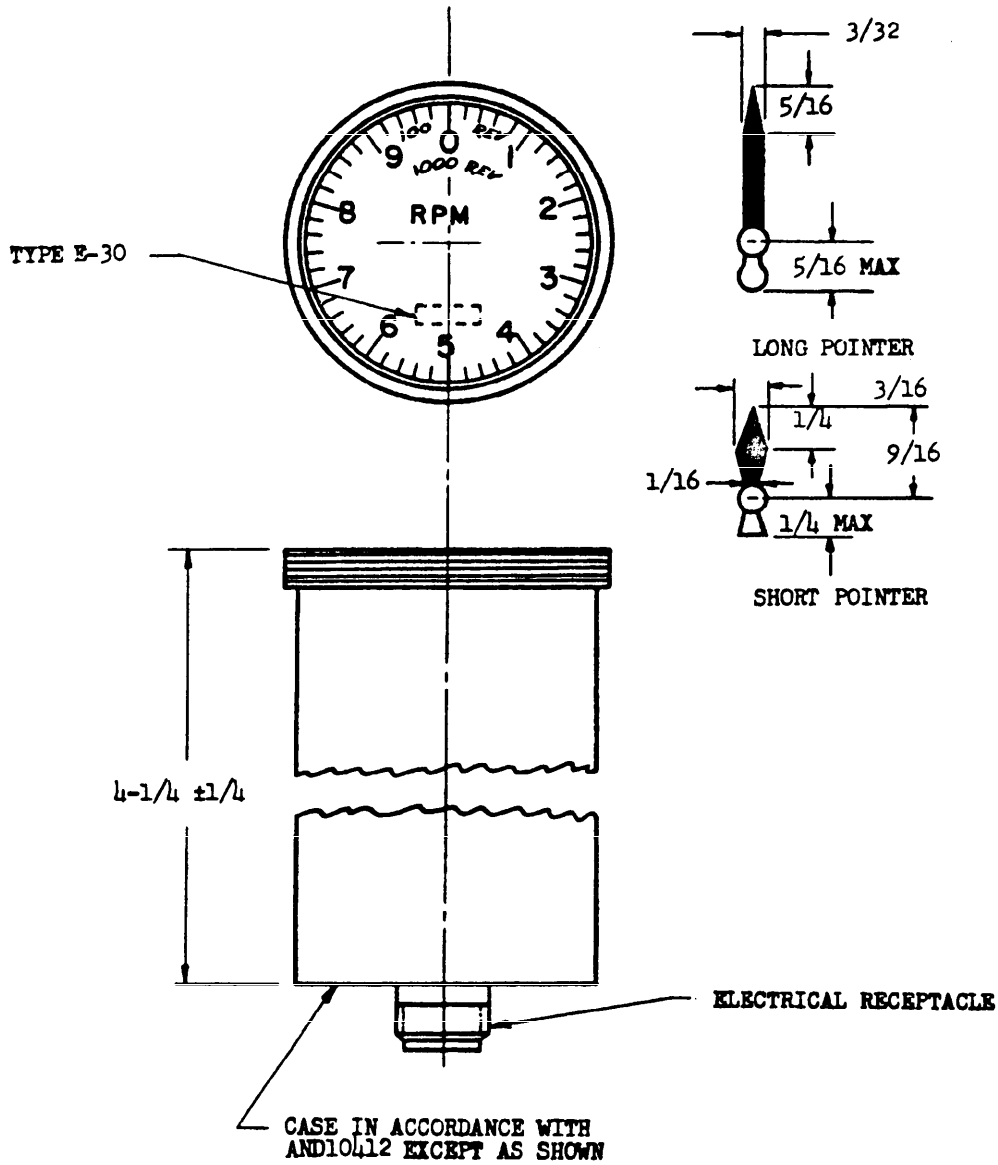
3.2.3.4 Scale Reading Overtravel.- The long pointer shall be free to rotate beyond an indication of 4,000 rpm, a distance equivalent to at least 50 rpm.

3.2.4 Markings.- The dimensions of the markings shall be as follows:

Marking	Height or Length ±0.016	Width of Line or Graduation ±0.005	Material or Finish
Numerals 0, 1, 2, 3, 4, 5, 6, 7, 8, and 9	0.156	0.025	
Graduations at 0, 1, 2, 3, 4, 5, 6, 7, 8, and 9	0.156	0.031	Fluorescent-Luminescent Material
20 RPM Graduations	0.094	0.020	
Lettering "RPM"	0.125	0.010	
Lettering "100 REV"	0.063	0.010	
Lettering "1000 REV"	0.063	0.010	
Shaded Portion of Pointers Type E-30	0.063	0.010	Durable Dull Black

3.2.5 Dial.- The dial shall have a minimum diameter of 1-3/4 inches when measured across the outside ends of the graduations and shall be in accordance with figure 1.

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DIMENSIONS IN INCHES UNLESS
OTHERWISE SPECIFIED TOLERANCE
ON: FRACTIONS - ±1/64

Figure 1 Dial and Pointers

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3.2.5.1 The marking "100 REV " shall be located on an arc whose radius is such that the top of the marking is 1/16 ±1/64 inch from the inner end of the 20 rpm graduations.

3.2.5.2 The marking "1000 REV " shall be so located that the top of the marking shall be located on an arc with a radius of 7/16 ±1/64 inch.

3.3 Weight.- The weight of each indicator shall not exceed 1 pound.

3.4 Identification of Product.-

3.4.1 Nameplate.- The nameplate shall be marked with the following information:

INDICATOR, TACHOMETER, ELECTRIC SENSITIVE
Type E-30
Specification MIL-I-7186
Manufacturer's Part No.
Manufacturer's Serial No.
Contract or Order No.
Stock No.
Manufacturer's Name or Trade-Mark
U S Property

3.4.2 Use of AN or MIL Designations.- AN or MIL designations shall not be applied to a product, except for Qualification test samples, nor referred to in correspondence or sales matter until notification has been received from the Aeronautical Standards Group that the product has been approved for aeronautical use, by both the Air Force and the Bureau of Aeronautics.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 General.- The tachometer indicator shall be subjected to inspection and tests in accordance with Specification MIL-I-7098, except that table I as specified herein shall be applicable in lieu of table I as listed in Specification MIL-I-7098.

TABLE I
Scale Errors

Generator Drive Shaft Speed	Frequency	Correct Indicated Speed	Scale Error Tolerance Room Temp.	Change in Scale Error at Specified Temperature (Maximum)			
RPM	Cycles Per Minute	RPM	RPM	RPM			
				-35°C	-54°C	+71°C	0°C
300	600	600 *	30	20	25	25	15
400	800	800	30	20	25	25	15
500	1000	1000 *	20	20	25	25	15
600	1200	1200	10	20	25	25	15
700	1400	1400 *	10	20	25	25	15
800	1600	1600	10	20	25	25	15
900	1800	1800 *	10	20	25	25	15
1000	2000	2000	10	20	25	25	15
1100	2200	2200 *	10	20	25	25	15
1200	2400	2400	10	20	25	25	15
1300	2600	2600	10	20	25	25	15
1400	2800	2800	10	20	30	30	20
1500	3000	3000 *	20	25	30	30	20
1600	3200	3200	30	30	40	40	25
1700	3400	3400	35	30	40	40	25
1800	3600	3600 *	40	35	50	50	30
1900	3800	3800	40	35	50	50	30
2000	4000	4000	40	35	50	50	30

* Test Points

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5. PREPARATION FOR DELIVERY

5.1 General.- The indicator shall be prepared for delivery in accordance with Specification MIL-I-7098.

5.2 Marking and Labeling.- The indicator container shall be marked with the following information:

INDICATOR, TACHOMETER, ELECTRIC, SENSITIVE, 2-INCH SIZE,
0-4,000 RPM
Type E-30
Specification MIL-I-7186
Manufacturer's Part No.
Manufacturer's Serial No.
Contract or Order No.
Stock No.
Date of Manufacture
Quantity
Manufacturer's Name or Trade-Mark
Name of Contractor (if not the same as manufacturer)

IF IN STORAGE AFTER (DATE)*, THE INDICATOR SHALL BE TESTED AND INSPECTED BEFORE ISSUE.

* This date shall be 18 months from the date of inspection and shall be inserted by the Inspector at the time of acceptance.

6. NOTES

6.1 General.- The instructions contained in Section 6 of Specification MIL-I-7098 are applicable to and form a part of this specification.

NOTICE: When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

Custodian:
Air Force

Other interest:
Navy - A
Army - T

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